

PRACTICAL 10

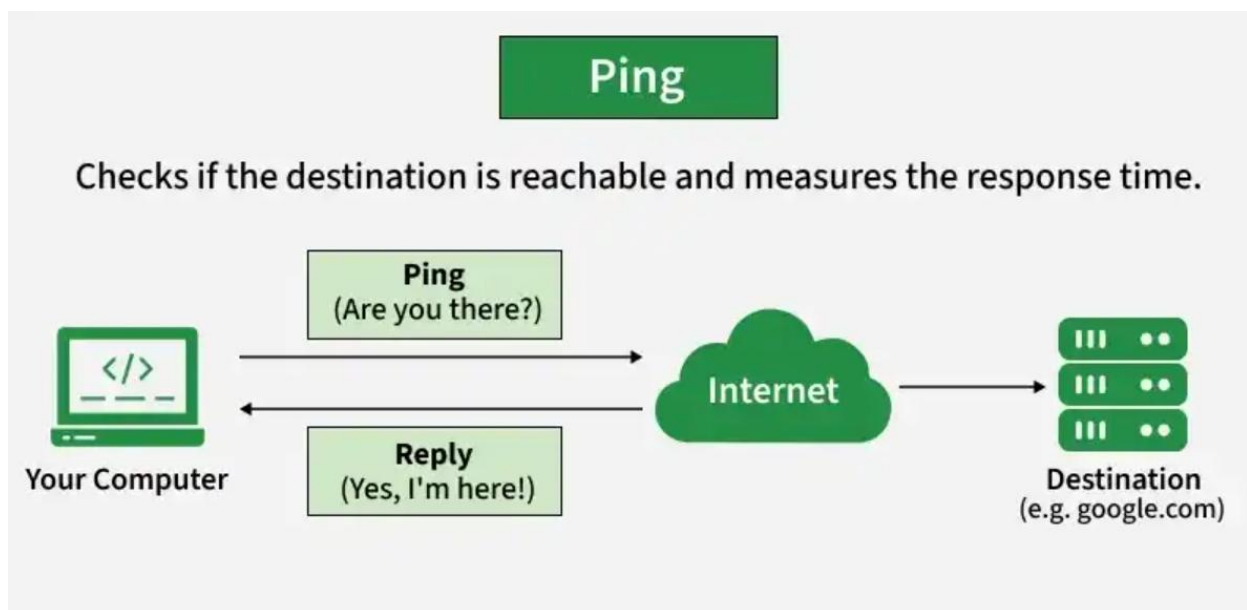
Aim : Run basic utilities and network commands : ipconfig, ping, tracert, netstat, pathping, route.

Input :

Network Troubleshooting is the process of finding and fixing problems in a computer network. It helps keep the network working properly and ensures good connectivity. When a network problem occurs, network administrators and IT professionals use tools such as Ping, Traceroute, and PathPing to identify the cause and resolve it.

Ping

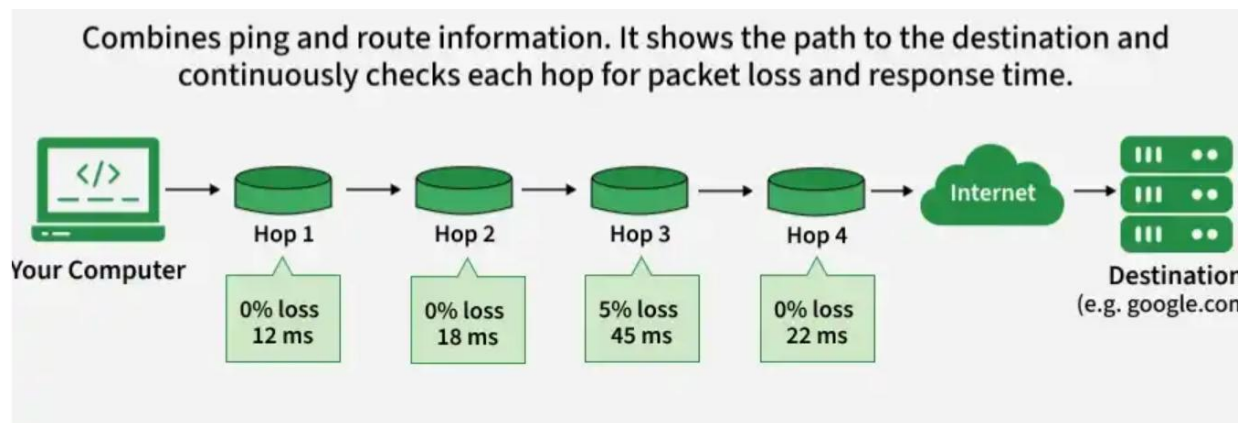
A Ping stands for Packet Internet Groper. It is a widely used command for identifying connectivity between two network connections. It uses Internet Control Message Protocol (ICMP) to send a request to the target host and wait for a response. It measures the round-trip time for data packets to travel from the source to the destination and back.



PathPing

PathPing command is a combination of ping and Traceroute. It sends request to each routers that comes between source and destination and compute result based on response from each router.

- It sends requests to each router and shows response time and packet loss, helping identify network problems.



Ipconfig

ipconfig stands for **Internet Protocol Configuration**. It is a Windows command used to display the IP configuration of a computer's network interfaces.

It provides information such as:

- IPv4 address
- IPv6 address
- Subnet mask
- Default gateway
- Network adapter information

1. ipconfig

Displays the IP configuration of the computer's network interfaces.



Your Computer

IP Address
Subnet Mask
Default Gateway
DNS Servers

```
C:\>ipconfig

Windows IP Configuration

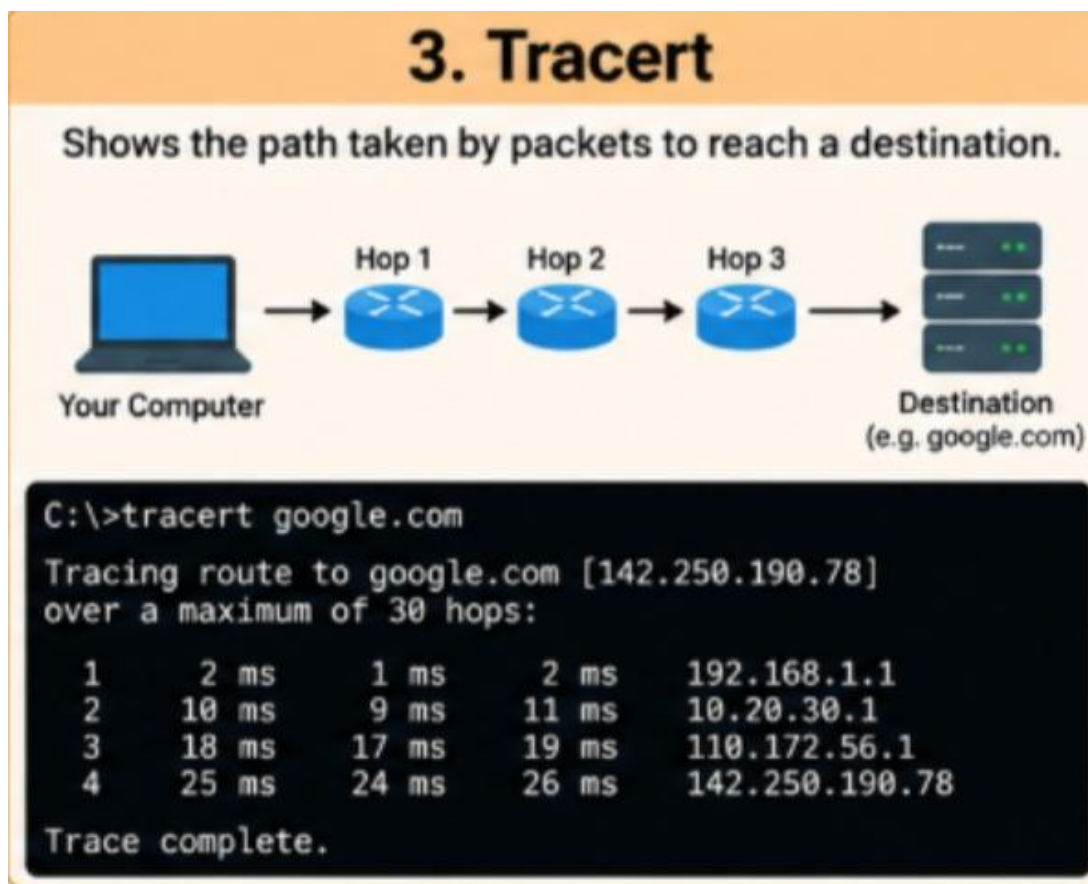
Wireless LAN adapter Wi-Fi:

    IPv4 Address. . . . . : 192.168.1.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
```

Tracert

tracert stands for **Trace Route**. It is a network diagnostic command used to determine the path taken by packets from the source computer to the destination.

It displays the different **network hops**, generally corresponding to routers or Layer-3 devices encountered along the path.



Netstat

netstat stands for **Network Statistics**. It is a command-line utility used to display information about active network connections and listening ports.

It can provide information about:

- TCP connections
- UDP endpoints
- Local addresses
- Foreign/remote addresses
- Connection states
- Listening ports
- Process IDs

4. Netstat

Displays active network connections, listening ports and process IDs.



```
C:\>netstat -ano
```

Proto	Local Address	Foreign Address	State	PID
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING	1200
TCP	192.168.1.15:52134	142.250.190.78:443	ESTABLISHED	4500
TCP	192.168.1.15:52135	142.250.190.78:443	ESTABLISHED	4600
UDP	0.0.0.0:5353	*:*		8700

Route

The route command is used to **display and modify the IP routing table** of a computer.

The routing table contains information that helps the operating system determine where packets should be forwarded.

